

Abdul Ghani Khan

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Summary

I'm a CS Graduate from FAST University (CGPA: 3.84/4.00) who genuinely enjoys building things — whether that's training a deep learning model on raw audio or wiring up a full-stack web app from scratch. Over the past couple of years I've worked on projects ranging from a content-based music recommendation system using PANNs to a multi-agent wellness assistant that ties together LLMs and vision models. I've done two internships at Synergy IT and one in AI/ML at CACF and currently serving as an intern at Code for Pakistan, so I'm comfortable on both the backend (.NET, MySQL, FastAPI) and the ML side (Python, scikit-learn). Outside of code, I've been on the Dean's List seven times and picked up six Gold Medals for highest semester SGPA.

Education

National University of Computer and Emerging Sciences (FAST), Peshawar Bachelor of Science in Computer Science — CGPA: 3.84 / 4.00	Aug 2022 – May 2026
Bahria College E-8 Naval Complex Islamabad Higher Secondary School Certification	2019 – 2021
Army Public School Westridge 3 Rawalpindi Secondary School Certification	2017 – 2019

Technical Skills

Languages	C++, Python, C#, JavaScript, TypeScript, SQL
Frameworks	Angular, Flutter, .NET, FastAPI, MySQL, PostgreSQL, Supabase, SQLAlchemy, Umbraco CMS, Scikit-learn, Pandas, Matplotlib, Streamlit, React, Cisco Packet Tracer, LEX/YACC
Core Areas	Machine Learning, Data Processing, Web APIs, Full Stack Development, OOP, Problem Solving

Experience

Technology Intern <i>Foundation for Civic Tech (Code for Pakistan)</i>	May 2026 – Jul 31, 2026
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- Contributing to the backend of **FT-AAFAT**, a centralized flood disaster management platform for government agencies, built with **FastAPI**, **PostgreSQL (Supabase)**, and **SQLAlchemy**.
- Designed core data models (Agencies, Users, Roles, Incidents, Resources, Volunteers, Notifications) and integrated **Supabase Auth** for role-based access control.
- Also contributing to **Rehbar**, a citizen-facing platform simplifying public access to government services and information.

AI/ML Intern <i>Central Asian Cellular Forum (CACF)</i>	Jul 2025 – Aug 2025
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- Completed AI Engineer Bootcamp covering AI and ML fundamentals.
- Developed a **Multi-Agent Wellness Assistant** integrating LLMs and vision models.
- Built agents for mental health, diet analysis, and exercise planning with database logging.
- Developed a Streamlit dashboard with chat interface, meal image analysis, and exercise tracking.
- Integrated Together.ai APIs (LLaMA 3.1 & 3.2) for text and vision tasks.

Full Stack Development Intern <i>Synergy IT</i>	Jul 2025 – Aug 2025
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- Developed backend services using **.NET Core** with database integration.
- Migrated legacy frameworks to .NET Core improving maintainability and performance.
- Contributed to development and integration tasks using **Umbraco CMS**.

Full Stack Development Intern <i>Synergy IT</i>	Jul 2024 – Aug 2024
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- Integrated .NET APIs with Angular frontend for blog platform functionality.
- Implemented full CRUD operations and backend validation.
- Managed MySQL database interactions and mailing features.

Teaching Assistant – Data Structures <i>FAST Peshawar (Assistant Professor Fazl-e-Basit)</i>	2025
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- Assisted in lab sessions, grading assignments, and mentoring students in Data Structures.

Teaching Assistant – Data Structures <i>FAST Peshawar (Lecturer Samin Ahmed)</i>	2025
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- Supported coursework delivery, evaluation, and student guidance in Data Structures.

Teaching Assistant – Digital Logic Design & Physics

FAST Peshawar (Assistant Professor Muhammad Asif Khan)

Spring & Fall 2024

- Assisted in coursework delivery, grading, lab activities, and student mentoring.

Projects

Content-Based Music Recommendation System (Final Year Project)

- Built deep learning music recommendation system using **PANNs** for audio feature extraction.
- Generated audio embeddings from raw music files and used cosine similarity for recommendations.
- Implemented preprocessing pipeline including sampling, spectrograms, and feature normalization.

Multi-Agent Wellness Assistant

- Designed multi-agent AI system with modules for mental health, diet tracking, and exercise planning.
- Implemented vision-based meal analysis and adaptive exercise planner.
- Built unified dashboard using **Streamlit** with text and audio input support.

Personal Blog Platform

- Full-stack application with blog creation, comments, mailing system, and admin dashboard.
- Technologies: Angular, .NET 8, MySQL.

Machine Learning Projects

- Customer segmentation using **K-Means** on Mall Customers dataset.
- Phishing email classification using **KNN** with Python and scikit-learn.
- Data analysis and visualization using Pandas and Matplotlib.

Power BI Dashboard – See Sight Tours

- Developed interactive dashboards using DAX to visualize revenue trends and business KPIs.

Leadership & Activities

Campus Director

Zindigi Prize

2024 – 2025

- Led campus team to organize entrepreneurship and innovation events.
- Managed event planning and industry engagement.

Vice President

FAST Trekking Society

2024 – 2025

- Organized trekking events and coordinated logistics and sponsorships.

Member

Google Developer Student Club (GDSC FAST)

2023 – 2024

- Participated in workshops on Android development and cloud technologies.

Participant

Young Leaders Connect 1.0

April 2025

- Represented FAST Peshawar in national leadership and innovation bootcamp.

Host

FAST Peshawar Graduation Ceremony

2024

- Hosted official graduation ceremony demonstrating public speaking and event management.

Awards

- **7× Dean's List Honoree**
- **6× Gold Medalist** – Highest SGPA per semester

Research Experience

Undergraduate Researcher – Quantum Information Lab

Quantum Information Lab (Supervised by Dr. Maqsood Khan) FAST University, Peshawar

Jan 2026 – Jun 30, 2026

- Working on a research project focused on noise-resilient quantum data provenance for synchrophasor networks
- Implementing quantum protocols using IBM Qiskit and analyzing noise in NISQ devices
- Applying quantum error correction techniques (e.g., Steane code) to improve system reliability
- Conducting simulations and evaluating system performance under realistic conditions
- Developing a simulation tool with interactive visualization for protocol analysis